

Matey Protitch

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(WhatsApp) Research repository: <https://zenodo.org/records/18888582>

Summary

Independent physics researcher and geospatial-data professional. Over nine months, single-handedly developed and published a nine-paper theoretical framework on the structure of the quantum vacuum (the “A-series,” Zenodo); one concept drew interest from **NASA NIAC**. Trained quantitative researcher (M.Sc. with graduate research in satellite and climate-data analysis), with international GIS and remote-sensing experience across the United States, China, Egypt, and Bulgaria, and a lifelong astronomy background (former editor of the popular-astronomy magazine *Andromeda*). Former U.S. permanent resident (eight years in the country); fluent English; available to relocate.

Independent Research — Physics (2025-present)

“Gravity Resonance” A-series — sole author, unaffiliated and self-funded. Nine papers in nine months applying standard physics — Compton-wavelength, de Broglie, and Kerr-geometry reasoning — to a proposed low-mass structure of the quantum vacuum (“granules”), yielding scales testable on existing instruments. One concept (nanoparticle resonator rings, A5) drew interest from **NASA NIAC** (200+ direct downloads of A5; ~450 across the full series). Independent AI assessments (Claude, Grok) of public LIGO O3, ARCADE 2, NANOGrav, and EDGES data reported features consistent with the framework; published 1/f frequency-noise spectra were independently reproduced from raw quantum-hardware records. (*Full publication list at the end; repository: Zenodo 10.5281/zenodo.18888582.*)

Education

- **M.Sc., Geography & GIS** — Indiana University of Pennsylvania, USA (2003). *Graduate Research Assistant*.
- **M.A., Geography (Ecology & Environmental Protection)** — Sofia University “St. Kliment Ohridski,” Bulgaria (1997).

Professional Experience

GIS Specialist & Chief Inspector — Sofia Municipal Inspectorate (Stolichen Inspektorat), Sofia, Bulgaria (Feb–Jun 2024) Produced GIS digital maps for the Inspectorate and the City of Sofia — bulky-waste collection zones with weekly schedules, central-district street-washing schedules, and locations of illegal dumpsites and municipal properties — published online (Google) and on the Inspectorate’s website. Initiated a city-wide GIS database: all streets of Sofia, river systems, and building footprints of Sofia, with updated GPS positions of waste containers — a first step toward an open-access municipal GIS on the model of major Western cities. Conducted field inspections of street and boulevard cleanliness and waste collection, investigated citizen reports, drafted official inspection protocols in the Inspectorate’s electronic system, and prepared official correspondence.

GIS Expert — Eurotechconsult Ltd (contractor to Sofia Municipality), Sofia, Bulgaria (2019–2020) European Investment Bank-financed waste-management optimization for Sofia Municipality. Built and maintained GIS databases; produced GPS- and satellite-based siting and optimization models for separate-collection infrastructure.

GIS Quality Assurance — Protrac, Cairo, Egypt (2006, ~3 months) Short-term contract: geo-database quality assurance (positional and attribute accuracy) and ISO 19115 metadata standards.

University Lecturer in GIS — China University of Geosciences, Wuhan, China (2003–2004) Taught ESRI ArcGIS / ArcView at China’s leading geoscience institution; conducted U.S.-market research for the MapGIS software product (Zondysoft Engineering).

English Lecturer — universities and schools, China (2003–2005) Hebei University, Shandong University of Engineering, Zhejiang Huawei School, and others.

GIS Technician (intern) — White Township Municipality, Indiana, PA, USA (2002) Land-use planning GIS; digitized aerial photography; DOQ/GPS correction of utility, road, and drainage layers; maps and presentations for municipal leadership.

Graduate Research Assistant — Indiana University of Pennsylvania, USA (2001–2002) Quantitative satellite and climate research: Landsat-7 analysis of Amazon (Rondônia) deforestation versus precipitation using Oak Ridge ABRACOS and GPCC datasets; SPSS stormwater-runoff prediction modeling.

Research Project Assistant — Institute of Floriculture, Sofia, Bulgaria (1995–1997) EU PHARE environmental-protection project: protected-species field surveys, habitat mapping, environmental analysis. **Co-author of three international conference papers.**

Science Communication & Community

- **Editor, astronomy magazine *Andromeda*; member, Astronomical Association of Sofia** (1997–2001). Public lectures for students, telescope-observation demonstrations, and summer astronomy camps.
- Editorial team, *Istina-Veritas*, the official newspaper of the Catholic Church in Bulgaria, Sofia (1993–2001).

Technical Skills

- **GIS / Remote Sensing:** ESRI ArcGIS Pro, ArcGIS 10, ArcGIS Online, QGIS, MapInfo; Trimble GPS; satellite imagery (Landsat, Micro MSI).
- **Programming / Data:** Python, Java, SPSS, VBA, HTML; AI-assisted development and customization (e.g., Docker); Microsoft Office.

Languages

English (fluent) · Bulgarian (native) · Russian (fluent comprehension, conversational) · Italian (good comprehension) · Norwegian (studied; basic).

Additional

- **2014–2022:** lived in the United States with extended family, serving as home-schooling educator for the family’s children. Former U.S. permanent resident (green card); EU / Bulgarian citizen; available to relocate.
- **2010–2014:** successfully treated for and fully recovered from leukemia.
- Participant in Bulgaria’s pro-democracy student movement during the fall of communism (1989–1991).

Publications — “Gravity Resonance” A-Series (Zenodo, 2025–2026)

Sole author. Full repository: <https://zenodo.org/records/18888582>

- **A9** — The Quantum Vacuum Granule: Three Complementary Mass Scales from the 1–5 Hz Universal Hum. doi:10.5281/zenodo.18888582
- **A8 (v2)** — Kerr Geometry and the Angular-Duality Wavelength: Mass Cancellation and Borderline Resonance from Nanoparticle to Galaxy. doi:10.5281/zenodo.19275058

- **A7 (v2)** — Imaginary Kerr Horizons in Ordinary Rotating Matter: A New Geometric Property and Its Possible Connection to the Universal 1–5 Hz Signal. doi:10.5281/zenodo.19320019
- **A6** — Hybrid van der Waals Graphene-Hydrogel-Mass-Seed Resonator for Asymmetric Mechanical Response and Electromechanical Output. doi:10.5281/zenodo.17684823
- **A5** — Quantum-Gravitational Double-Resonance Propulsion Rings: Spacecraft-Scale Architecture and Physical Basis. doi:10.5281/zenodo.17634226
- **A4** — Imaginary Kerr Horizon in Every Rotating Body Below One Solar Mass: A Universal Quantum-Gravity Boundary. doi:10.5281/zenodo.17619734
- **A3** — De Broglie-Kerr Ergosphere Resonance in Rotating Coherent Systems: Listening to the Horizon. doi:10.5281/zenodo.17594467
- **A2** — Soft-Matter Toroid for Quantum-Gravitational Resonance: Toward a Tabletop Realization of de Broglie-Schwarzschild Symmetry. doi:10.5281/zenodo.17541112
- **A1b** — Toroid Models for Quantum-Gravitational Resonance via de Broglie-Schwarzschild Symmetry. doi:10.5281/zenodo.16449277

Independent AI assessments of the series (Grok / xAI; Claude Opus and Claude Sonnet / Anthropic) are linked from the repository.